AYMANE AIT DADS

AI Research Intern (Embodied Robotics) | PyTorch · Computer Vision · Multimodal Perception

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Hiring Team - AI Research Intern (Embodied Robotics)
Sony Electronics Singapore (Pentas Vision / Sony Semiconductor Solutions Corporation) · Singapore

Dear Hiring Team,

Your requirement for research in **multimodal perception** and **embodied AI** maps directly to the work I do: fine-tuning large vision models to perceive and distinguish real-world signals under distribution shift. At EURECOM, I ranked **#1 on the class leaderboard** by fine-tuning **CLIP ViT-L/14** (428M parameters) on 250K images spanning 25+ generator types - designing test-time augmentation strategies specifically to handle out-of-distribution inputs, the same generalization challenge that makes sensor validation in real-world robotics hard.

To support Pentas Vision's goal of **validating new sensors and algorithms** in embodied tasks, I bring two directly relevant proof points. First, I built a **transformer autoencoder** for unsupervised fault detection on industrial audio achieving **AUC 0.80** with no labeled anomalies - a protocol matching real deployment constraints where ground truth is unavailable. Second, I ran structured **ablation experiments** across unfreezing depth (4/6/8 transformer layers), learning rate schedules, and augmentation strategies, producing documented findings on generalization trade-offs - exactly the experimental rigor needed to validate perception algorithms across **simulation and real-world** environments.

Pentas Vision sits at the intersection of Sony Semiconductor Solutions' imaging sensor hardware and embodied intelligence software - a combination where perception research has direct physical stakes. Contributing to that pipeline, where my vision and signal processing work feeds into a physical robot's ability to interpret its environment, is the specific problem I want to work on. I would welcome the chance to discuss how my background fits your current research agenda.

Sincerely,

AYMANE AIT DADS